DO NOW, AO1 REVISION

Resistance and I–V characteristics (RP)

Electricity
Paper 1, Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 8 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the two measurements needed to find the resistance of a component.**	The potential difference across it and the current through it	☐
2	**Wire length and resistance: name the independent variable.**	Length of the wire	☐
3	**Wire length and resistance: name the dependent variable.**	Resistance of the wire	☐
4 ★	**Wire length and resistance: name three control variables.**	The temperature, thickness and material of the wire	☐
5 ★	**Describe how the length of a wire affects its resistance.**	The resistance increases (it is directly proportional to length).	☐
6 ★	**In the wire-length practical, the fixed crocodile clip is at the 1 cm mark, but every length is read from 0 cm. Name the type of error this causes.**	A zero error (a systematic error): every length is 1 cm too long.	☐
7 ★	**Explain why the potential difference is kept small when measuring the resistance of a wire.**	So the current is small and the wire does not heat up.	☐
8	**What does it mean for two resistors to be in series?**	They are connected one after the other, on a single path.	☐
9	**What does it mean for two resistors to be in parallel?**	They are on separate branches.	☐
10 ★	**Describe how to calculate the total resistance of two resistors in series.**	Add the two resistances: $R_{total} = R_1 + R_2$.	☐

	QUESTION	ANSWER	✓
11	**A lamp, a power supply and a variable resistor are in series. State two ways to increase the potential difference across the lamp.**	Increase the supply potential difference. Decrease the resistance of the variable resistor.	☐
12 ★	**Finding the I–V characteristic of a component: name the independent and dependent variables.**	Independent: potential difference across the component Dependent: current through it	☐
13 ★	**Describe the graph of resistance against length for a wire.**	A straight line through the origin.	☐

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